Physics

Ans 1. The beam will be reflected at infinity as the image is formed there.

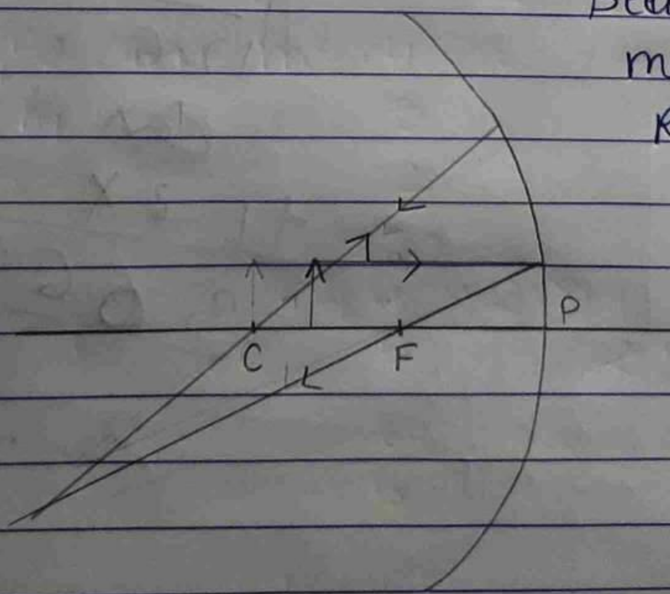
Ans 2. R = 50 cm So $R = \frac{f}{2} \Rightarrow 50 = \frac{f}{2}$

$f = 50 \times 2 \Rightarrow 100$

So focal length is 100 cm

It is a Convex spherical mirror.

Ans 3.



Between F and C
magnified and
Real and
inverted

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Ans 4. $h_1 = 2.0$ cm
$f = 10$ cm
$u = -15$ cm
$v = ?$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{v} + \frac{1}{-15} = \frac{1}{10}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{10} - \frac{1}{-15}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{10} + \frac{1}{15}$$

5	10, 15
2	2, 3
	3

$$\begin{array}{r} 15 \\ \times 2 \\ \hline 30 \end{array}$$

$$\Rightarrow \frac{1}{v} = \frac{3+2}{30}$$

$$\Rightarrow \frac{1}{v} = \frac{5}{30}$$

$$\Rightarrow v = \frac{\cancel{30}\,6}{\cancel{5}\,1} \Rightarrow v = 6 \text{ cm}$$

$$\frac{h_2}{h_1} = \frac{-V}{u}$$

$$\Rightarrow \frac{h_2}{2} = \frac{-6}{-15}$$

$$\Rightarrow -15h_2 = -12$$

$$\Rightarrow h_2 = \frac{\cancel{+12}\ 4}{\cancel{+15}\ 5}$$

$$h_2 = \frac{4}{5}$$

$$m = \frac{h_2}{h_1} \Rightarrow m = \frac{\frac{4}{5}}{2}$$

$$= \frac{\cancel{4}^{\,2}}{5} \times \cancel{2} \otimes \frac{1}{\cancel{2}\ 1}$$

$$\cancel{= \frac{8}{5}}$$

$$m = \frac{-V}{u}$$

$$m = \frac{+\cancel{6}\ 2}{+\cancel{15}\ 5}$$

$$\Rightarrow \frac{2}{5} = 0.4 \text{ cms}$$

$$5\,\overline{)\,20\,}\ 0.4$$
$$-20$$
$$\times$$

nature = Virtual and erect
position = 6 cm
size = 0.4 cm